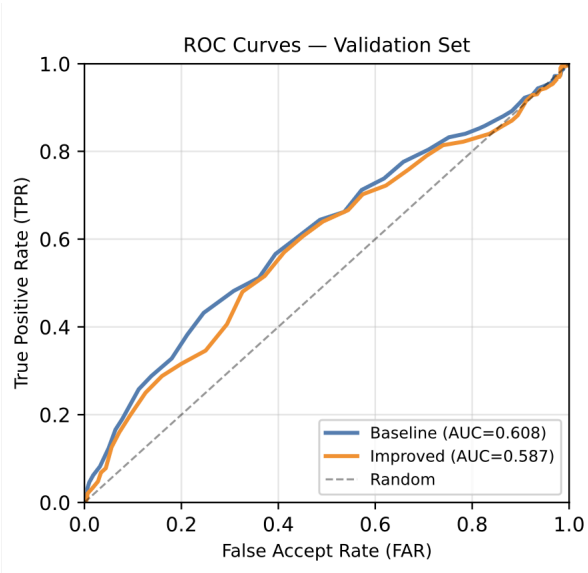
# MSML/MSAI 605 — Milestone 2 Report

## Face Verification on LFW

This report presents the Milestone 2 evaluation of a cosine-similarity-based face verification pipeline built on the LFW dataset. The system takes deterministic pairs of face images and produces a similarity score; a threshold converts this into a same-person vs. different-person decision. Pairs are split by identity (70% train, 15% val, 15% test). The threshold was selected on the validation set only, using the rule of maximizing balanced accuracy, before any test-set evaluation.

## Setup and tracked runs

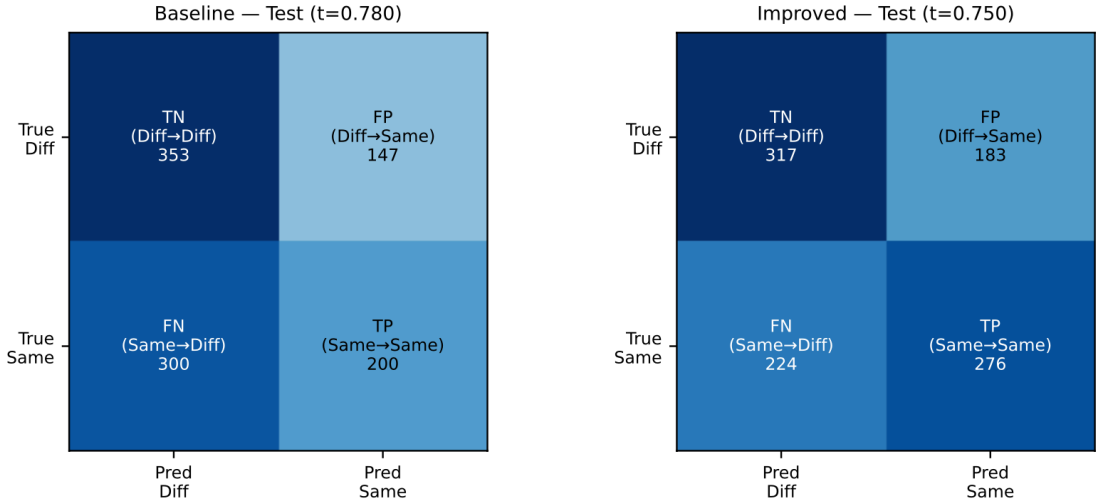
Five runs were tracked using MLflow. Runs 1–3 cover the baseline pipeline: a threshold sweep on validation, a selected-threshold evaluation, and final test reporting. Runs 4–5 repeat this process after a data-centric improvement. All runs record the git commit hash, config version, split, threshold, and metrics.



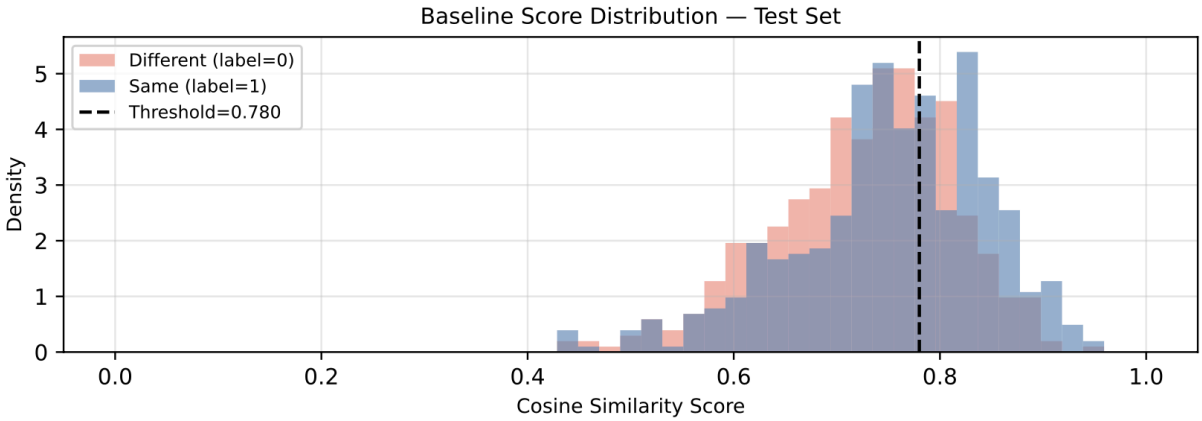

*ROC Curves on the validation set. Baseline AUC = 0.608, Improved AUC = 0.587.*

## Threshold selection

The operating threshold was chosen by sweeping values from 0.0 to 1.0 on the validation split and selecting the value that maximized balanced accuracy. This rule was fixed before any test evaluation. The baseline selected threshold is **0.780** (balanced accuracy on val = 0.553). The improved pipeline selected **0.750** (balanced accuracy on val = 0.593).




*Left: Baseline test set (t=0.780). Right: Improved test set (t=0.750).*




*Cosine similarity score distributions for same-identity (blue) and different-identity (red) pairs on the test set. The threshold at 0.780 sits in a region of heavy overlap, explaining the high error rates.*

## Run Summary & Error Analysis

| Run ID | Split | Data | Threshold | BalAcc | FAR | FRR | AUC | Note |
|---|---|---|---|---|---|---|---|---|
| baseline_val_sweep | val | baseline | sweep | — | — | — | 0.608 | Baseline threshold sweep on validation set |
| baseline_val_selected | val | baseline | 0.780 | 0.593 | 0.246 | 0.568 | — | Selected threshold using rule 'maximize balanced accuracy' |
| baseline_test_final | test | baseline | 0.780 | 0.553 | 0.294 | 0.600 | — | Baseline final evaluation on test set |
| improved_val_sweep | val | improved | sweep | — | — | — | 0.587 | Improved data (max 5 images/identity) threshold sweep |
| improved_test_final | test | improved | 0.750 | 0.593 | 0.366 | 0.448 | — | Improved data final evaluation on test set |

*All 5 tracked MLflow runs with split, threshold, balanced accuracy, FAR, FRR, and AUC.*

## Error Slice 1 — Rare Identities (≤3 images):

Positive pairs where the identity has 3 or fewer images in the full dataset.

**Size:** 308 pairs.

**False Reject Rate:** 0.6006 (185 false rejects).

**Hypothesis:** With only 1–2 possible image combinations per identity, pose and lighting variation is entirely uncontrolled. These pairs are structurally harder — the two images may be the only ones available, taken in very different conditions, pushing cosine similarity below the threshold even for the same person.

**Suggested fix:** Filter identities with fewer than 4 images from the positive pair pool, or apply a quality proxy to flag high-variation pairs.

## Error Slice 2 — Hard Negatives (top 25% similarity among negatives):

Different-identity pairs whose cosine similarity falls in the top 25% of all negative pair scores (cutoff = 0.7885).

**Size:** 125 pairs.

**False Accept Rate:** 1.000 — every pair in this slice is misclassified.

**Hypothesis:** These pairs have high pixel-level similarity likely due to shared lighting conditions, similar backgrounds, or shared demographic appearance features. Pixel-based cosine similarity has no mechanism to separate identity-irrelevant visual similarity from true identity matches.

**Suggested fix:** Replace pixel cosine similarity with a learned face embedding (e.g. FaceNet or ArcFace) that is trained to be invariant to appearance and sensitive to identity.

## Data-Centric Change:

**Change:** Capped each identity to a maximum of 5 images during pair generation (baseline had no cap; one identity had ~530 images).

**Motivation:** Without capping, high-frequency identities dominate the positive pair pool, making evaluation unrepresentative of realistic verification conditions.

**Results:** Balanced accuracy improved from 0.553 → 0.593. FRR dropped from 0.600 → 0.448 (the system catches more same-identity pairs). FAR increased from 0.294 → 0.366 (more hard negatives accepted — an expected tradeoff at the lower threshold of 0.750).

## LEARNING:

The most important lesson from this iteration is that the pipeline's failure modes are not random — they are systematic and predictable. Rare-identity pairs and visually similar negatives account for a disproportionate share of errors. Addressing these with better representations (learned embeddings) and better data curation (identity capping, quality filtering) are the clearest paths to improvement in Milestone 3.